JEE–MAIN EXAMINATION – JANUARY 2026

(HELD ON WEDNESDAY 21st JANUARY 2026) **TIME : 3:00 PM TO 6:00 PM**

MATHEMATICS

TEST PAPER WITH ANSWER

SECTION-A

1. The positive integer n, for which the solutions of the equation

$x(x+2) + (x+2)(x+4) + ...+(x+2n-2)(x+2n) = \frac{8n}{3}$

are two consecutive even integers, is :-

(1) 3 (2) 6

(3) 12 (4) 9

2. Let $f : \mathbf{R} \to \mathbf{R}$ be a twice differentiable function such that $f''(x) > 0$ for all $x \in \mathbf{R}$ and $f'(a-1) = 0$, where a is real number. Let $g(x) = f(\tan^2 x - 2\tan x + a)$, $0 < x < \frac{\pi}{2}$.

Consider the following two statements :

(I) g is increasing in $\left(0, \frac{\pi}{4}\right)$

(II) g is decreasing in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

Then,

(1) Neither (I) nor (II) is True

(2) Only (II) is True

(3) Only (I) is True

(4) Both (I) and (II) are True

3. Let $f(x) = x^3 + x^2 f'(1) + 2x f''(2) + f'''(3)$, $x \in R$. Then the value of $f'(5)$ is :

(1) $\frac{62}{5}$ (2) $\frac{657}{5}$

(3) $\frac{2}{5}$ (4) $\frac{117}{5}$

4. In the line $\alpha x + 4y = \sqrt{7}$, where $\alpha \in R$, touches the ellipse $3x^2 + 4y^2 = 1$ at the point P in the first quadrant, then one of the focal distances of P is :

(1) $\frac{1}{\sqrt{3}} - \frac{1}{2\sqrt{11}}$ (2) $\frac{1}{\sqrt{3}} + \frac{1}{2\sqrt{5}}$

(3) $\frac{1}{\sqrt{3}} - \frac{1}{2\sqrt{5}}$ (4) $\frac{1}{\sqrt{3}} + \frac{1}{2\sqrt{7}}$

5. Let $y^2 = 12x$ be the parabola with its vertex at O. Let P be a point on the parabola and A be a point on the x-axis such that $\angle OPA = 90°$. Then the locus of the centroid of such triangles OPA is :

(1) $y^2 - 6x + 4 = 0$

(2) $y^2 - 9x + 6 = 0$

(3) $y^2 - 2x + 8 = 0$

(4) $y^2 - 4x + 8 = 0$

6. Let one end of a focal chord of the parabola $y^2 = 16x$ be (16, 16). If $P(\alpha, \beta)$ divides this focal chord internally in the ratio 5 : 2, then the minimum value of $\alpha + \beta$ is equal to :

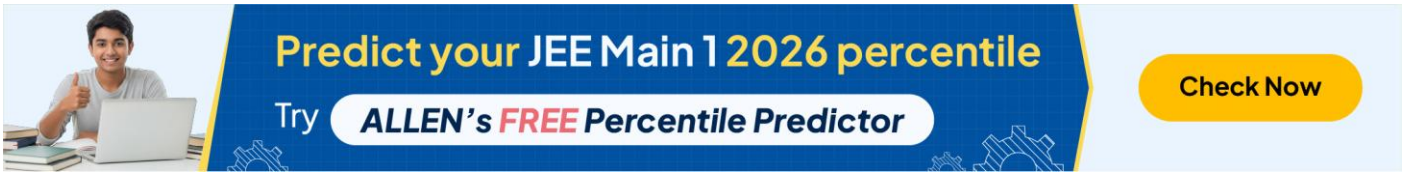
(1) 22 (2) 7

(3) 5 (4) 16

7. Let the line L pass through the point (–3, 5, 2) and make equal angles with the positive coordinate axes. If the distance of L from the point (–2, r, 1) is $\sqrt{\frac{14}{3}}$, then the sum of all possible values of r is :

(1) 12 (2) 16

(3) 6 (4) 10



8. Let the line L_1 be parallel to the vector $-3\hat{i}+2\hat{j}+4\hat{k}$ and pass through the point (2, 6, 7) and the line L_2 be parallel to the vector $2\hat{i}+\hat{j}+3\hat{k}$ and pass through the point (4, 3, 5). If the line L_3 is parallel to the vector $-3\hat{i}+5\hat{j}+16\hat{k}$ and intersects the lines L_1 and L_2 at the points C and D, respectively, then $\left|\overrightarrow{CD}\right|^2$ is equal to :

(1) 171 (2) 290

(3) 312 (4) 89

9. Let α and β be the roots of equation $x^2+2ax+(3a+10)=0$ such that $\alpha<1<\beta$. Then the set of all possible values of a is :

(1) $\left(-\infty,\frac{-11}{5}\right)\cup(5,\infty)$

(2) $(-\infty,-2)\cup(5,\infty)$

(3) $(-\infty,-3)$

(4) $\left(-\infty,\frac{-11}{5}\right)$

10. A random variable X takes values 0, 1, 2, 3 with probabilities $\frac{2a+1}{30},\frac{8a-1}{30},\frac{4a+1}{30}$, b respectively, where a, b $\in$ **R**. Let μ and σ respectively be the mean and standard deviation of X such that $\sigma^2+\mu^2=2$. Then $\frac{a}{b}$ is equal to :

(1) 30 (2) 3

(3) 60 (4) 12

11. If the area of the region $\{(x, y) : 1-2x\le y\le 4-x^2, x\ge 0, y\ge 0\}$ is $\frac{\alpha}{\beta}$, $\alpha, \beta, \in N$, gcd $(\alpha, \beta) = 1$, then the value of $(\alpha+\beta)$ is :

(1) 73 (2) 85

(3) 91 (4) 67

12. Let $a_1, \frac{a_2}{2}, \frac{a_3}{2^2}, \ldots, \frac{a_{10}}{2^9}$ be a G.P. of common ratio $\frac{1}{\sqrt{2}}$. If $a_1+a_2+\ldots+a_{10}=62$, then a_1 is equal to :

(1) $2\left(\sqrt{2}-1\right)$ (2) $2-\sqrt{2}$

(3) $\sqrt{2}-1$ (4) $2(2-\sqrt{2})$

13. Let $A=\{x : |x^2-10|\le 6\}$ and $B=\{x : |x-2|>1\}$. Then

(1) $A\cup B=(-\infty,1]\cup(2,\infty)$

(2) $A-B=[2,3)$

(3) $A\cap B=[-4,-2]\cup[3,4]$

(4) $B-A=(-\infty,-4)\cup(-2,1)\cup(4,\infty)$

14. For the matrices $A=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$ and $B=\begin{bmatrix}-29 & 49\\ -13 & 18\end{bmatrix}$, if $(A^{15}+B)\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$, then among the following which one is true?

(1) x = 5, y = 7

(2) x = 18, y = 11

(3) x = 11, y = 2

(4) x = 16, y = 3

15. For a triangle ABC, let $\vec{p}=\overrightarrow{BC}, \vec{q}=\overrightarrow{CA}$ and $\vec{r}=\overrightarrow{BA}$. If $|\vec{p}|=2\sqrt{3}, |\vec{q}|=2$ and $\cos\theta=\frac{1}{\sqrt{3}}$, where θ is the angle between $\vec{P}$ and $\vec{q}$, then $\left|\vec{p}\times\left(\vec{q}-3\vec{r}\right)\right|^2+3\left|\vec{r}\right|^2$ is equal to:

(1) 340 (2) 220

(3) 410 (4) 200



16. Let y = y(x) be the solution of the differential equation $\sec x\frac{dy}{dx}-2y = 2 + 3\sin x$, $x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, $y(0)=-\frac{7}{4}$. Then $y\left(\frac{\pi}{6}\right)$ is equal to:

(1) $-\frac{5}{2}$ (2) $-\frac{5}{4}$

(3) $-3\sqrt{3}-7$ (4) $-3\sqrt{2}-7$

17. Let A = {2, 3, 5, 7, 9}. Let R be the relation on A defined by x Ry if and only if $2x \le 3y$. Let ℓ be the number of elements in R, and m be the minimum number of elements required to be added in R to make it a symmetric relation. Then ℓ + m is equal to:

(1) 23 (2) 25

(3) 21 (4) 27

18. If the system of equations

$3x + y + 4z = 3$

$2x + \alpha y - z = -3$

$X + 2y + z = 4$

has no solution, then the value of α is equal to:

(1) 19 (2) 4

(3) 13 (4) 23

19. Let z be the complex number satisfying $|z-5|\le 3$ and having maximum positive principal argument.

Then $34\left|\frac{5z-12}{5iz+16}\right|^2$ is equal to:

(1) 16 (2) 12

(3) 26 (4) 20

20. The largest $n\in N$, for which 7^n divides 101!, is :

(1) 16 (2) 18

(3) 15 (4) 19

SECTION-B

21. Let [·] denote the greatest integer function and $f(x) = \lim\limits_{n\to\infty}\frac{1}{n^3}\sum\limits_{k=1}^{n}\left[\frac{k^2}{3^x}\right]$. Then $12\sum\limits_{j=1}^{\infty}f(j)$ is equal to _____.

22. If $\int\limits_0^1 4\cot^{-1}(1-2x+4x^2)dx = a\tan^{-1}(2) - b\log_e(5)$, where $a, b\in\mathbf{N}$, then $(2a + b)$ is equal to _____.

23. Let the maximum value of $(\sin^{-1}x)^2 + (\cos^{-1}x)^2$ for $x\in\left[-\frac{\sqrt{3}}{2},\frac{1}{\sqrt{2}}\right]$ be $\frac{m}{n}\pi^2$, where gcd (m, n) = 1.

Then m + n is equal to _____.

24. If

$$\left(\frac{1}{{}^{15}C_0}+\frac{1}{{}^{15}C_1}\right)\left(\frac{1}{{}^{15}C_1}+\frac{1}{{}^{15}C_2}\right)\cdots\left(\frac{1}{{}^{15}C_{12}}+\frac{1}{{}^{15}C_{13}}\right) = \frac{\alpha^{13}}{{}^{14}C_0\,{}^{14}C_1\cdots{}^{14}C_{12}},$$

then 30α is equal to _____.

25. If P is a point on the circle $x^2 + y^2 = 4$, Q is a point on the straight line $5x + y + 2 = 0$ and $x - y + 1 = 0$ is the perpendicular bisector of PQ, then 13 times the sum of abscissa of all such point P is _____.



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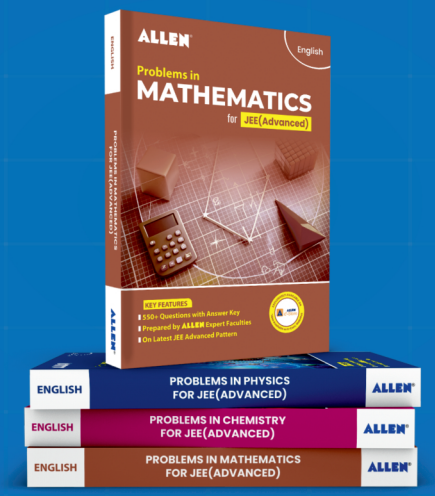


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